

WROCLAW STRACHOWICE, Poland

WMO: 124240

Lat: 51.1032N

Lon: 16.8998E

Elev: 397

StdP: 14.49

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	13.5	2.8	6.4	9.1	8.1	8.4	14.7	25.3	41.2	22.6	42.2	4.2	120	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	87.7	68.8	84.2	67.2	80.9	65.9	70.8	83.4	69.1	80.5	67.4	77.5	7.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.1	75.3	65.0	93.8	73.6	63.7	89.5	72.3	34.9	83.7	33.4	80.7	32.0	77.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.8	15.6	2.3	93.4	7.1	3.4	-2.8	95.9	-6.9	97.9	-10.9	99.8	-16.1	102.2		
			1.7	74.0	6.9	1.7	-3.3	75.3	-7.3	76.3	-11.2	77.2	-16.2	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	31.2	33.8	39.7	49.2	57.4	63.6	67.3	66.6	58.2	49.6	40.9	33.9
	DBStd	14.67	9.50	8.89	7.56	7.25	6.52	6.26	5.59	5.55	5.98	7.15	7.23	8.58
	HDD50	2359	584	454	325	100	11	0	0	0	5	97	282	501
	HDD65	5995	1049	873	784	474	248	99	38	47	215	479	724	965
	CDD50	2129	0	1	6	76	241	409	538	514	252	83	8	1
	CDD65	291	0	0	0	0	13	58	111	96	12	1	0	0
	CDH74	3071	0	0	0	30	187	612	1147	938	151	6	0	0
	CDH80	908	0	0	0	2	30	172	393	278	33	0	0	0
Wind	WSAvg	7.0	8.1	8.0	8.1	7.2	6.6	6.5	6.1	5.7	6.2	6.7	7.3	8.0
Precipitation	PrecAvg	22.7	1.1	1.0	1.2	1.4	2.4	2.8	3.5	2.8	1.9	1.5	1.5	1.3
	PrecMax	30.5	3.8	1.9	2.9	3.1	5.5	7.3	9.9	9.0	4.3	5.0	4.1	3.9
	PrecMin	15.4	0.2	0.1	0.2	0.2	0.2	0.4	0.4	0.2	0.3	0.1	0.0	0.2
	PrecStd	3.7	0.6	0.5	0.6	0.7	1.3	1.4	1.9	1.9	1.1	1.0	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	57.1	66.0	78.1	83.9	90.3	92.8	92.2	84.5	74.2	62.2	54.5
		MCWB	48.2	48.9	52.4	60.4	65.6	71.0	69.4	70.6	66.0	61.5	54.5	49.4
	2%	DB	50.1	52.7	60.5	71.9	79.1	84.7	88.1	86.3	78.1	69.0	58.0	51.1
		MCWB	45.5	46.8	49.7	57.0	63.5	68.0	68.3	68.7	63.4	59.2	52.4	46.8
	5%	DB	46.5	49.5	55.6	67.4	75.1	80.8	84.4	82.7	74.0	64.8	54.8	48.3
		MCWB	42.8	44.6	47.9	54.6	61.8	65.7	67.1	66.9	61.9	57.3	50.1	44.8
	10%	DB	43.4	46.2	51.9	63.2	71.3	77.0	80.6	79.2	70.1	60.9	51.7	45.3
		MCWB	40.3	42.3	45.8	52.6	59.3	63.7	66.3	65.4	60.3	55.3	48.0	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	50.2	54.3	61.9	67.4	73.1	72.9	72.6	67.9	63.3	55.8	50.1
		MCDB	53.0	55.3	63.5	74.8	79.3	86.0	86.4	86.7	80.6	71.8	60.8	54.0
	2%	WB	45.9	47.3	50.8	58.8	65.1	69.8	70.7	70.4	65.2	60.1	52.8	47.2
		MCDB	49.4	51.7	58.0	69.4	76.0	82.2	83.0	82.9	74.8	67.0	57.2	50.6
	5%	WB	43.0	45.0	48.6	55.8	62.9	67.2	69.0	68.6	63.3	58.0	50.6	44.9
		MCDB	46.1	49.2	54.8	65.2	73.3	77.9	80.2	79.3	71.7	63.8	54.2	47.9
	10%	WB	40.4	42.2	46.1	53.4	60.8	65.2	67.4	66.9	61.3	55.8	48.2	42.4
		MCDB	43.1	46.0	51.3	61.9	68.9	74.4	77.2	76.2	68.7	60.5	51.2	45.1
Mean Daily Temperature Range	5% DB	MDBR	10.2	12.1	15.1	18.9	19.5	19.1	19.3	20.1	17.9	15.7	11.9	10.1
		MCDBR	12.6	16.6	23.8	27.4	26.6	26.3	27.1	26.4	25.1	22.5	17.2	12.5
		MCWBR	9.9	11.9	15.2	14.9	13.7	12.1	10.8	11.3	12.7	13.7	12.3	9.7
	5% WB	MCDBR	12.2	15.5	21.4	24.7	24.0	23.8	22.9	22.9	22.0	20.6	16.1	12.1
MCWBR		9.8	11.5	14.4	14.1	13.1	12.0	10.6	10.7	11.9	13.0	12.0	9.8	
Clear-Sky Solar Irradiance	taub	0.317	0.343	0.383	0.430	0.424	0.421	0.444	0.434	0.396	0.370	0.344	0.311	
	taud	2.321	2.299	2.223	2.141	2.205	2.246	2.195	2.234	2.311	2.346	2.366	2.360	
	Ebn at Noon	220	245	255	256	264	266	256	252	249	232	207	203	
	Edh at Noon	21	28	36	44	43	42	43	40	33	26	20	17	
All-Sky Solar Radiation	RadAvg	247	470	815	1311	1608	1693	1599	1429	1017	582	282	185	
	RadStd	43	83	114	173	185	160	184	128	139	81	40	22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.12	N/A	N/A	+1.64	+1.22	+1.26	N/A	-347	+212	+72
(47) Regional (35 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+36

Nomenclature: See separate page